AZURE

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1.What is Microsoft Azure?

Microsoft Azure is a cloud computing platform provided by Microsoft that allows us to store data, run applications, create databases, and process data over the internet.

What is Cloud Computing?

Cloud computing means using computing resources such as servers, storage, databases, and networking over the internet instead of managing everything on our own computers.

Examples of cloud platforms:

- Microsoft Azure
- AWS
- Google Cloud Platform (GCP)

Types of cloud services:

- **Infrastructure as a Service (IaaS):** Rents out virtual machines, storage, and networking hardware while you manage the operating system and applications.
- **Platform as a Service (PaaS):** Provides a managed environment for developers to build and run software without worrying about underlying hardware or OS maintenance.
- **Software as a Service (SaaS):** Delivers ready-to-use cloud applications over the internet (such as Microsoft 365).

What is an Azure Subscription?

An Azure subscription is a logical container and agreement with Microsoft that gives you authenticated access to deploy, manage, and use cloud resources like virtual machines and databases. It acts as both a **billing boundary** to track and aggregate costs and an **administrative boundary** for role-based access control (RBAC).

Core Functions

- **Billing Unit:** Generates invoices and tracks usage costs for all resources provisioned inside it.
- **Access Control (RBAC):** Lets you delegate permissions and assign security roles to different teams or users.
- **Resource Container:** Holds logical groups of cloud infrastructure, such as storage accounts, networks, and virtual machines.

Common Subscription Types

- **Free Account:** Includes limited-time credits (such as $200 for the first 30 days) and select free services for 12 months to test the platform.
- **Pay-As-You-Go:** Charges you monthly based on actual resource consumption with no upfront cost or long-term commitment.
- **Enterprise Agreement (EA):** Built for large organizations that commit to a fixed amount of resource usage upfront in exchange for volume-based discounts.
- **Dev/Test:** Provides discounted rates for non-production workloads, development, and testing.

What is a Resource Group?

A **resource group** is a logical container in cloud platforms like Microsoft Azure that holds related cloud services—such as virtual machines, databases, web apps, and storage accounts—so you can manage them as a single unit.

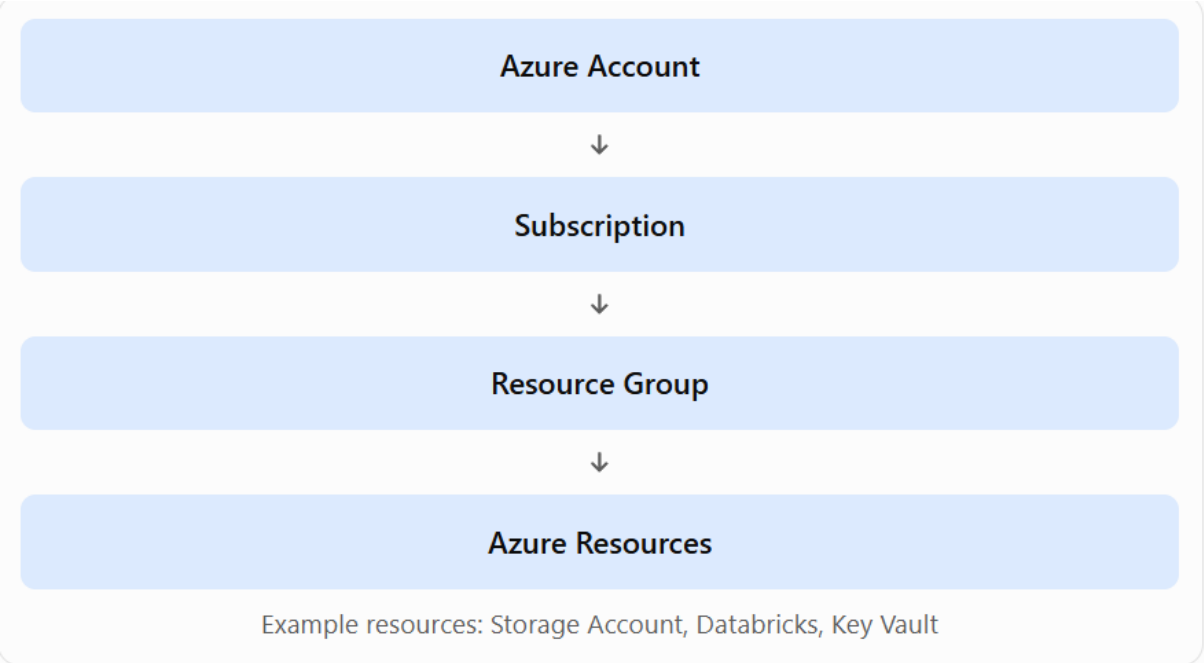
Example: A data engineering project might have a resource group containing:

- Azure Storage Account
- Azure Databricks workspace
- Azure Key Vault

What is a Azure Resource?

An Azure resource is a manageable item that is available through the cloud. Think of an Azure service like a software class, and a resource as an active, running instance of that class. Common examples include virtual machines, storage accounts, web apps, SQL databases, and virtual networks.

Azure Hierarchy



2.What is an Azure Storage Account?

An **Azure storage account** is a secure, unified cloud container that groups and provides access to all of your Microsoft Azure Storage data services.

Core Data Services Included

A single storage account provides a unique namespace and acts as a wrapper for these core data services:

- **Blob Storage:** Object storage for massive amounts of unstructured data like images, videos, logs, and documents.
- **Azure Files:** Managed file shares accessible via industry-standard SMB or NFS protocols.
- **Queue Storage:** A messaging service for reliable communication and asynchronous processing between application components.
- **Table Storage:** A NoSQL key-value store designed for structured or semi-structured data.

What is ADLS Gen2?

Azure Data Lake Storage Gen2 (ADLS Gen2) is Microsoft's enterprise big data analytics storage solution built directly on top of Azure Blob Storage. It combines the low-cost, high-scale storage benefits of blob storage with a true hierarchical file system.

Core Features

- **Hierarchical Namespace (HNS):** Unlike standard flat blob storage, ADLS Gen2 organizes data into true directories and subfolders. This makes folder operations (like renaming or deleting large directories) fast metadata changes rather than slow file-by-file copies.
- **Hadoop Compatibility:** It natively supports the Hadoop Distributed File System (HDFS) via the Azure Blob File System (`abfs://` or `abfss://`) driver.
- **Fine-Grained Security:** It integrates Azure Role-Based Access Control (Azure RBAC) with POSIX-style Access Control Lists (ACLs) for granular folder- and file-level permissions.

Difference between Blob Storage and ADLS Gen2?

Feature	Azure Blob Storage	ADLS Gen2
Purpose	General object storage	Big data analytics
Organization	Containers and blobs	Containers, directories, and files
Hierarchical namespace	Optional	Enabled for ADLS Gen2
Access control	Azure RBAC and other storage mechanisms	Azure RBAC plus file/directory ACLs
Use case	Images, backups, documents	Data engineering and analytics

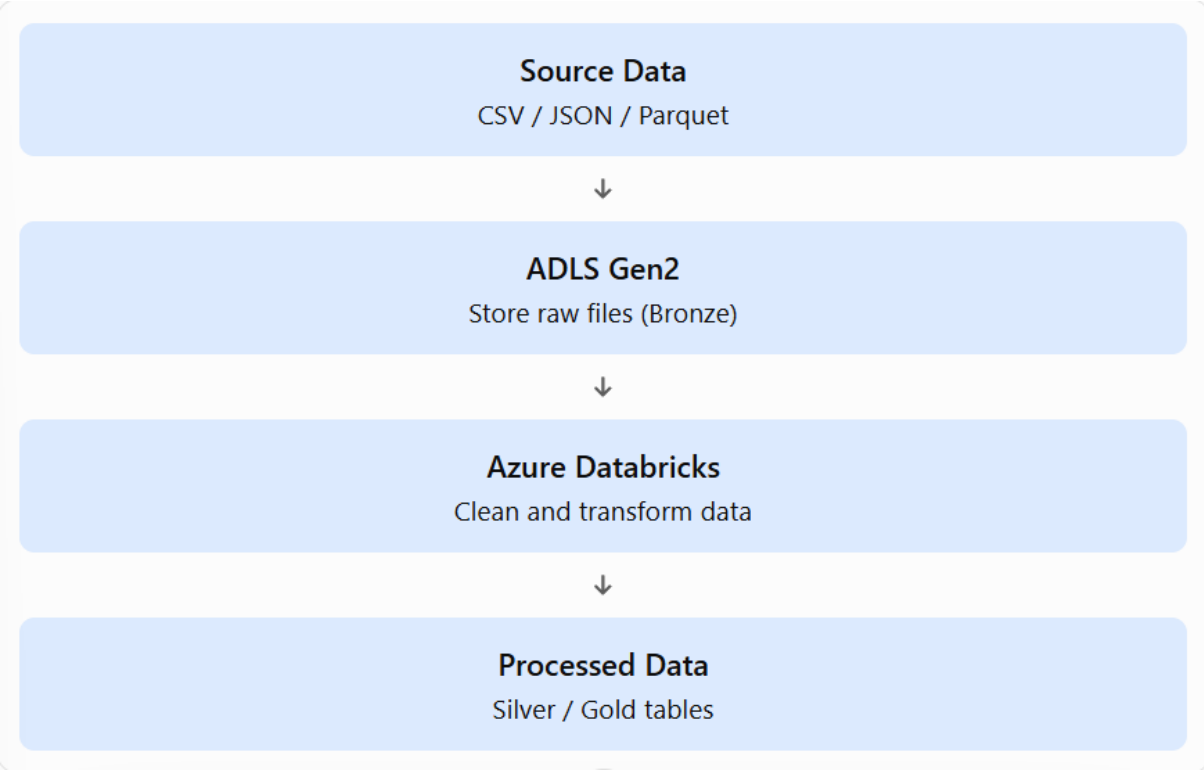
What are Storage Access Tiers?

Storage access tiers are cloud data management levels that balance storage costs against access frequency. Cloud providers like Azure Storage Overview allow you to pay more for high-speed, frequent access or pay less for storage while accepting higher fees when you read or retrieve the data.

Main Types of Access Tiers

- **Hot Tier**: Optimized for data that you access or modify often. It features the highest storage costs but the lowest access and transaction costs.
- **Cool Tier**: Best for data you rarely access that must stay stored for at least 30 days. It offers lower storage costs and higher access costs than the hot tier.
- **Cold Tier**: Built for rarely accessed data that still requires fast online retrieval and should be kept for at least 90 days. It sits between cool and archive in cost.

How is ADLS Gen2 used in Data Engineering?



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1.Containers, Hierarchical Namespace, and Storage Access:

1.1.What is a Container in Azure?

A container in Azure is a lightweight, standalone, and portable software package that bundles application code, libraries, and dependencies so it can run reliably across any computing environment.

1.2.What is Hierarchical Namespace (HNS)?

A **Hierarchical Namespace (HNS)** is a data management feature that organizes files and objects into a tree-like structure of real directories and subdirectories, replacing traditional flat object storage models.

1.3.What is Storage Access?

Storage Access most commonly refers to the Storage Access API in web development, or permission models governing file and cloud storage systems.

Method	Purpose
Azure RBAC	Grants permissions through roles
ACLs	Controls access to specific files and directories
SAS token	Provides limited access for a specified time and permissions
Storage account key	Provides broad access to storage

1.4.What are Read, Write, and Execute Permissions?

The Three Basic Permissions

- **Read (r)**
  - **On Files:** Allows you to open and view the contents of a file (e.g., reading a text document or script).
  - **On Directories:** Allows you to list the file and folder names inside that directory (using a command like `ls`).
- **Write (w)**
  - **On Files:** Allows you to modify, edit, overwrite, or delete the file's contents.
  - **On Directories:** Allows you to add new files, delete existing files, or rename items inside the directory. *(Note: To actually delete or modify a file inside a directory, you typically need execute permission on that directory as well).*
- **Execute (x)**

- **On Files:** Allows you to run the file as a program, script, or binary.
- **On Directories:** Allows you to traverse, enter, or access the directory (e.g., using `cd`) and inspect metadata of the files inside.

2.Microsoft Entra ID and Managed Identities:

2.1.What is Microsoft Entra ID?

Microsoft Entra ID is Microsoft's cloud-based identity and access management service. It helps authenticate users and applications accessing Azure resources.

Example: When you sign in to the Azure Portal using your Microsoft account, Microsoft Entra ID helps verify your identity.

Key points:

- Manages users and groups.
- Supports authentication and single sign-on.
- Helps applications securely access resources.
- Works with Azure role-based access control (RBAC).

2.2. Authentication vs Authorization

Authentication	Authorization
Verifies who you are	Determines what you can access
Example: Signing in to Azure	Example: Permission to read an ADLS file

2.3. What is a Managed Identity?

A Managed Identity is an identity managed by Azure that allows an Azure service to authenticate to supported services without you having to store credentials in your application code.

Example: An Azure Data Factory pipeline needs to access an ADLS Gen2 storage account. You can configure its managed identity and grant the necessary permissions.

Without a managed identity, you might need to manage credentials manually. With a managed identity, Azure handles the identity credentials.

2.4. Types of Managed Identities

System-assigned identity

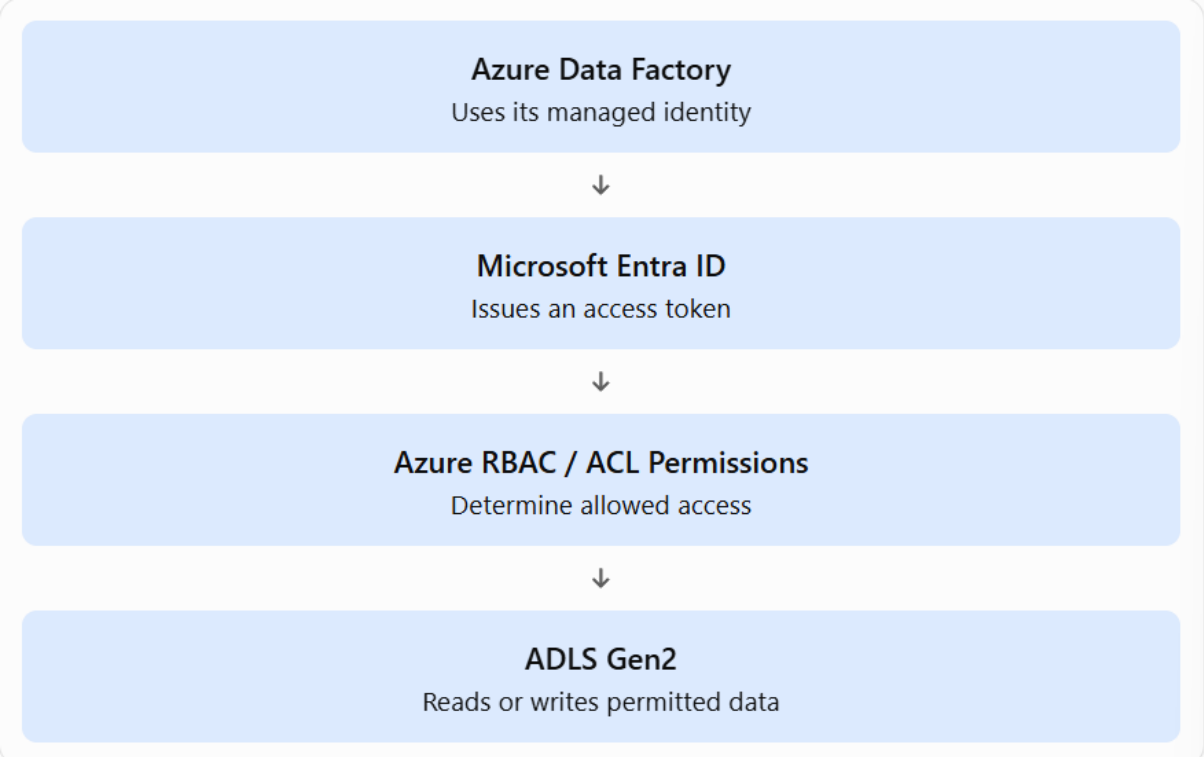
- Created as part of a specific Azure resource.
- Its lifecycle is tied to that resource.

- Deleted when the associated resource is deleted.

User-assigned identity

- Created as a separate Azure resource.
- Can be assigned to multiple supported Azure resources.
- Has an independent lifecycle.

Process of Managed Identity Access ADLS Gen2?



3.Secret Management and Key Rotation in Azure:

3.1.What is Secret Management?

Secret management is the process of securely storing, accessing, updating, and protecting sensitive information used by applications.

Examples of secrets:

- Database passwords
- API keys
- Access tokens
- Connection strings
- Certificates and private keys

3.2. What is Azure Key Vault?

Azure Key Vault is a cloud service used to securely store and manage secrets, encryption keys, and certificates.

Main uses:

- Store database passwords and API keys.
- Manage encryption keys.
- Store and manage certificates.
- Control access to sensitive information.
- Support secret rotation and auditing.

Example: An Azure Data Factory pipeline needs a database password. The password can be stored in Azure Key Vault and retrieved securely when the pipeline runs, using a supported integration and appropriate permissions.

3.3. What is Key Rotation?

Key rotation is the process of replacing an existing key or credential with a new one periodically or when required to improve security.

Example:

1. An application uses a storage account key.
2. The organization generates a replacement key.
3. The application is updated to use the new key.
4. The old key is retired after confirming everything works.

3.4. Secrets vs Keys vs Certificates

Item	Purpose	Example
Secret	Stores sensitive values	Database password
Key	Used for cryptographic operations or access, depending on the key type	Encryption key
Certificate	Establishes identity or supports secure communication	TLS certificate

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1.Managed Identities and RBAC Role Assignments in Azure:

1.1. What is a Managed Identity

A Managed Identity is an identity managed by Microsoft Azure that allows an Azure service to authenticate to other supported services without storing credentials in your code.

Types of Managed Identities

- **System-Assigned**: Tied directly to a single Azure resource (such as a Virtual Machine or App Service). If you delete the resource, Azure automatically deletes the identity.
- **User-Assigned**: Created as a standalone resource in Azure. You can assign it to one or more Azure resources, and its lifecycle is separate from them

System-assigned	User-assigned
Created for a specific Azure resource	Created as a separate Azure resource
Lifecycle is tied to that resource	Has an independent lifecycle
Deleted when the resource is deleted	Can be reused by multiple supported resources

1.2.What is Azure RBAC:

Azure RBAC Azure role-based access control is an authorization system built on Azure Resource Manager that manages who has access to Azure resources, what actions they can perform on those resources, and which areas or scopes they can access.

Core Components

Azure RBAC operates through three primary elements that form a **role assignment**:

- **Security Principal:** An object representing a user, group, service principal (application), or managed identity requesting access.
- **Role Definition:** A collection of permissions and actions (such as `read`, `write`, or `delete`) that define what a principal can do. Azure provides over 70 built-in roles as well as custom roles.
- **Scope:** The set of resources to which the access applies. Scopes are hierarchical and can be applied at the **Management Group**, **Subscription**, **Resource Group**, or individual **Resource** level. Permissions flow downward to lower levels.

Four Fundamental Built-in Roles

- **Owner:** Full access to manage all resources, including the ability to delegate access to other users.
- **Contributor:** Can create and manage all types of Azure resources, but cannot grant access to other users.
- **Reader:** Can view existing Azure resources but cannot make any changes.
- **User Access Administrator:** Lets you manage user access to Azure resources (often paired with assigning specific roles)

1.3.What is a Role Assignment:

Role assignment is the process of linking a specific set of permissions and responsibilities (a role) to a user, group, or service principal within an IT, cloud, or operational system. It acts as the core mechanism for enforcing security models like Role-Based Access Control (RBAC).

Key Components

- **Principal:** The entity receiving access (such as a user account, security group, or managed identity).
- **Role Definition:** The bundle or collection of allowed and excluded actions, policies, and privileges.
- **Scope:** The boundary or level where the permissions apply (such as an individual file, a database, a resource group, or an entire cloud subscription).

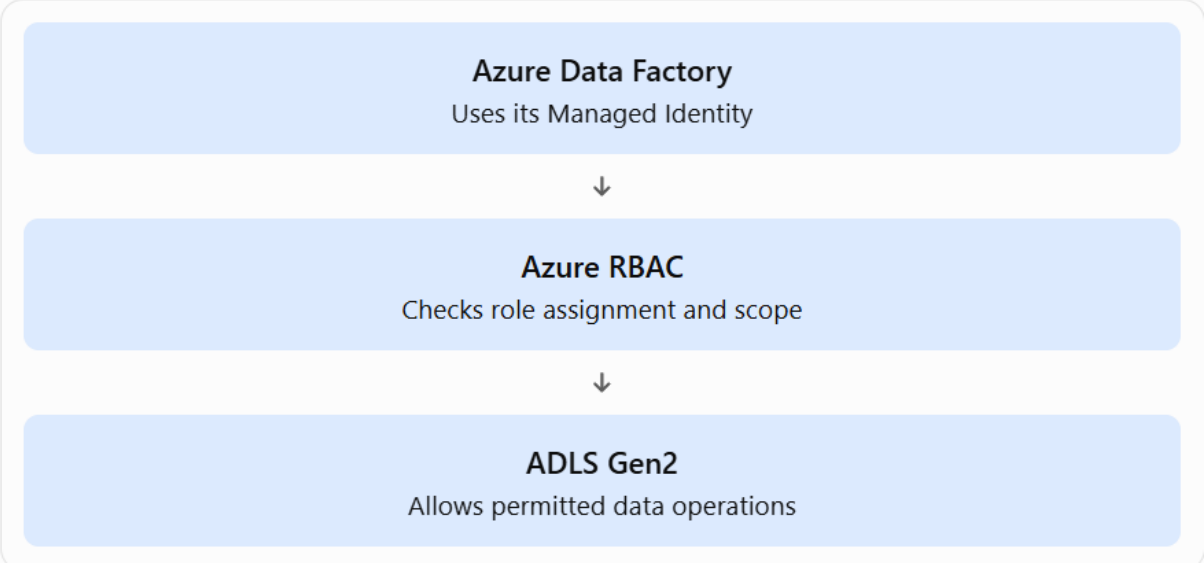
1.4.What is the Storage Blob Data Contributor role:

The **Storage Blob Data Contributor** is a built-in Azure role that allows a security principal (user, group, service principal, or managed identity) to read, write, and delete Azure Storage blob containers and blobs.

Key Permissions & Capabilities

- **Data Actions (`dataActions`):** Read, write (add, overwrite, move), and delete blobs and containers within the assigned scope.
- **Container Management:** Create, delete, and read blob containers.
- **User Delegation Key:** Can request a user delegation key to generate User Delegation SAS (Shared Access Signature) tokens when assigned at the storage account scope or higher.

WorkFlow:



2.Azure Key Vault Fundamentals — Secrets and Keys:

2.1.What is Azure Key Vault?

Azure Key Vault is a cloud service that securely stores and manages secrets, cryptographic keys, and certificates.

Example: If your data pipeline needs a database password, you can store it in Key Vault instead of writing it directly in your Python code.

2.2. What are Secrets:

Secrets in Azure are confidential pieces of data—such as passwords, API keys, database connection strings, or private keys—that applications and services use to authenticate or access resources.

Examples:

- Database passwords
- API keys
- Connection strings
- Access tokens

2.3.What are Keys:

Keys in Azure Key Vault are cryptographic objects used to encrypt and decrypt data (such as data at rest) or to sign and verify digital signatures.

Types of Keys

- **RSA Keys**: Asymmetric keys used for encryption, decryption, and signing/verifying. Supported sizes include 2048-bit, 3072-bit, and 4096-bit.
- **Elliptic Curve (EC) Keys**: Asymmetric keys used primarily for digital signatures and ECDH key agreement. Supported curves include P-256, P-384, P-521, and secp256k1/P-256K.
- **Symmetric Keys (`oct-HSM` / AES)**: Hardware-protected symmetric keys (128-bit, 192-bit, and 256-bit) supported in Azure Key Vault Premium and Managed HSM.

Examples:

- Encryption keys
- RSA keys
- Elliptic Curve keys

Difference: A secret stores a sensitive value, while a cryptographic key performs security-related operations.

2.4.What are Certificates:

A certificate helps establish an identity or secure communication between systems.

Example: A TLS certificate helps a website establish a secure HTTPS connection.

Azure Key Vault can manage certificates along with secrets and keys.

3.Azure Key Vault — RBAC vs Access Policies

3.1. What is Azure RBAC in Key Vault:

Azure role-based access control (Azure RBAC) in Key Vault is an authorization system that manages who can access and perform operations on your key vaults, keys, secrets, and certificates.

3.2.What are Key Vault Access Policies:

Azure Key Vault Access Policies are a legacy authorization system that determines whether a security principal (such as a user, group, service principal, or managed identity) has permission to perform data plane operations on keys, secrets, and certificates inside an Azure Key Vault.

Difference Between RBAC and Access Policies

Azure RBAC	Access Policies
Uses Azure role assignments	Uses explicit Key Vault access policies

Supports centralized role management	Manages data permissions through the vault's access policies
Can assign built-in or custom roles	Selects specific secret, key, and certificate permissions
Supports broader Azure authorization patterns	Uses the legacy Key Vault access-policy permission model

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